

6(a)(i)	anaphase ;	1
6(a)(ii)	<i>any two from:</i> growth ; R growth of cells repair of tissues (by replacement of cells) ; replacement of, old / dead /damaged / worn out, cells ; R repair of cells asexual reproduction ; production of genetically identical daughter cells ;	2

Question	Answer	Marks
6(b)	any four from: as concentration of Paclitaxel increases, the ratio of cells in anaphase to those in metaphase reduces / there are a greater proportion of cells in metaphase than in anaphase ; as the concentration of Paclitaxel increases, the percentage of cells in mitosis is increasing ; use of data to support a described trend ; <i>idea that</i> as the concentration increases, more cells, stop in metaphase / spend more time in metaphase / fewer cells are able to move into anaphase ; suggested mechanism for halt in metaphase ; e.g. centromeres do not divide prevents spindle fibres shortening prevents movement of chromatids to opposite poles (because sister chromatids still held together) cells do not pass the (metaphase) checkpoint AVP ;	4